

AI-Powered Smart Waste Intelligence Platform

Project Overview

This platform combines computer vision, explainable AI, sustainability analytics, and environmental intelligence systems to improve recycling awareness and waste management efficiency.

AI Performance

EfficientNet-B0 achieved:

- Validation Accuracy: 98.5%
- Strong explainability localization
- Stable real-time inference
- Edge-AI deployment suitability

Sustainability Impact

The system estimates:

- CO2 reduction impact
- Recycling efficiency
- Waste distribution trends
- Environmental awareness metrics

Research Contributions

- Multi-model benchmarking
- Explainable AI experimentation
- Human-centered AI design

- Sustainability-oriented analytics
- Real-time intelligent classification

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